SPECIFICATIONS PLATFORM

# Phase 3

## Component Inventory & Design System

**Deliverables 7–8 of 12**

Prepared for **Moses Mkhumbuza**
Project **Malls Specs — reusable specification portal**
Contents **18 specimen plates · 59 components · 74 verified colour pairings**
Ships with **tokens.css · tailwind.tokens.js · tokens.ts · CONTRAST-REPORT.md**
Date **7 August 2026**
Status **Awaiting approval to proceed to Phase 4**

CONFIRMED IN PHASE 2

Split-pane variant A + B — panel widens to 58% above 1440px; attachments, revision history and related specifications render full-width beneath the split.
Landing opens with a compact hero above the browse surface.
Status vocabulary — Draft, In review, Published, Deprecated, Superseded.
Metadata is dynamic key/value pairs, authored per specification or inherited from a category template.

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# 1. How this system is built

Three rules govern everything that follows. They are what make the system reusable rather than merely tidy.

## 1.1 Tokens are named by role, never by value

There is no token called "navy" or "blue-500". There is fg-muted, border-control, accent-subtle, brand-surface. A component asks for the role it needs and receives whatever the active theme binds to it. This is the single decision that makes dark mode cost one stylesheet block instead of a parallel component tree, and it is why plate S18 shows two themes rendered from identical markup.

## 1.2 Contrast is solved, not chosen

Every neutral step is derived by binary search against a contrast target rather than picked as an opacity that looked right. fg-muted is whatever composite of the ink over the surface reaches 7.2:1; fg-subtle is whatever reaches 4.65:1. The same function that produces those values produces the verification report that ships with the tokens, so the document and the code cannot drift apart.

This matters because it is exactly how the reference system went wrong. Its neutrals were chosen as opacities — 55% and 35% — which look reasonable and fail measurably.

## 1.3 The reference system’s character is preserved deliberately

This is a refinement, not a redesign, and the confirmed direction is to keep the split-pane and the visual language. So the navy-and-accent relationship, the two-curve motion vocabulary, the label-plus-flexing-rule section header, the bulleted specification row, the navigation selection marker and the reference-code chip all survive essentially unchanged. What changes is what was measurably broken: contrast, type floor, focus, semantics and state coverage.

> **WHAT THE NUMBERS SAY**
>
> Reference system audit: 7 of 12 representative colour pairings fail WCAG 2.1 AA.
>
> New palette: 0 failures across 74 pairings, light and dark.
>
> Component inventory: 59 components — 6 retained unchanged, 8 extended, 8 rebuilt, 1 consolidated, 1 derived, 35 new.
>
> The new components are not embellishment. They are the status, tags, revision history, attachments, related specifications, anchor navigation, print view, bookmarks, keyboard navigation, dark mode and the four system states your brief requires and the reference system does not have.

## 2. Foundations

The token layer. Everything in Section 4 is assembled from these values and references nothing else.

## S1 Colour — light mode

### SURFACES

| Aa | Aa | Aa | Aa | Aa |
|---|---|---|---|---|
| --color-bg | --color-bg-subtle | --color-bg-muted | --color-bg-strong | --color-bg-inverse |
| #FFFFFF 16.48:1 | #F5F7FC 15.37:1 | #EEF1F8 14.57:1 | #E3E8F4 13.43:1 | #16294C 14.43:1 |

### FOREGROUNDS

| | | | | |
|---|---|---|---|---|
| --color-fg | --color-fg-muted | --color-fg-subtle | --color-brand | --color-accent-fg |
| #0D1E3F 16.48:1 | #4B5870 7.17:1 | #6A7589 4.65:1 | #1B3769 11.69:1 | #1A4FD6 6.7:1 |

### ACCENT

| Aa | Aa | Aa | Aa | |
|---|---|---|---|---|
| --color-accent | --color-accent-hover | --color-accent-active | --color-accent-subtle | --color-accent-border |
| #2861FD 4.98:1 | #1A4FD6 6.7:1 | #1540B0 8.78:1 | #EDF2FF 5.98:1 | #B9CCFF 1.6:1 |

### BORDERS

| | | | |
|---|---|---|---|
| --color-border | --color-border-strong | --color-border-control | --color-border-focus |
| #E7E8EC 1.22:1 | #CFD2D9 1.51:1 | #8C94A4 3.05:1 | #2861FD 4.98:1 |

### SEMANTIC · FOREGROUND ON SUBTLE BACKGROUND

| Aa info | Aa success | Aa warning | Aa danger | Aa neutral |
|---|---|---|---|---|
| --color-info-* | --color-success-* | --color-warning-* | --color-danger-* | --color-neutral-* |
| 4.44:1 | 4.82:1 | 5.34:1 | 5.72:1 | 7.24:1 |

### SPECIFICATION STATUS · CONFIRMED VOCABULARY

| DRAFT | IN REVIEW | PUBLISHED | DEPRECATED | SUPERSEDED |
|---|---|---|---|---|
| status.draft | status.review | status.published | status.deprecated | status.superseded |
| DRAFT | IN REVIEW | PUBLISHED | DEPRECATED | SUPERSEDED |

1 Every neutral is the ink colour composited over the surface, so the palette stays tonally unified — the single best property of the reference system, preserved exactly.

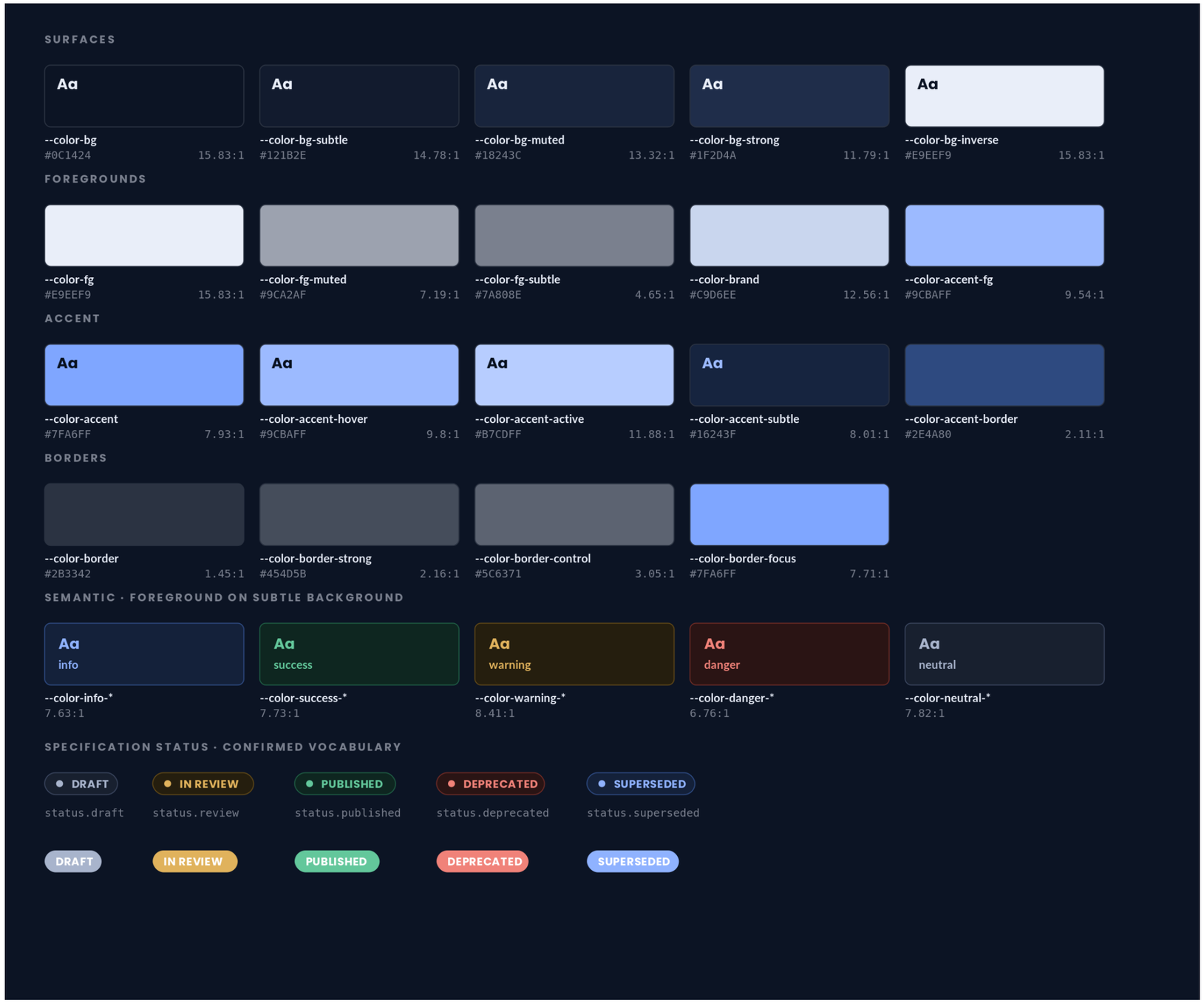
2 The neutral steps are solved for a contrast target rather than picked as an opacity. fg-muted lands at 7.2:1 and fg-subtle at 4.65:1, both against the base surface.

3 border-control is a separate token from border. Input and button boundaries must reach 3:1 under WCAG 1.4.11; decorative hairlines have no such requirement and stay subtle.

4 Status uses the semantic scale rather than its own colours, so a new status needs no new palette.

5 Soft badges are the default. Solid badges are reserved for the specification action bar, where the badge sits against a tinted surface.

## S2 Colour — dark mode




1 Dark mode is derived from the same token names, not a separate palette. Components reference --color-fg and never a literal, so nothing needs a dark variant.

2 Accent lightens to #7FA6FF because the light accent fails against a dark surface. Foreground-on-accent inverts to near-black.

3 The neutral ramp is solved against the dark base to the same targets — 7.2:1 and 4.65:1 — so the tonal rhythm matches light mode rather than merely inverting.

4 Semantic backgrounds are deep tints of their hue rather than dark greys, so status remains identifiable at a glance.

5 Theme is set by a data-theme attribute, with prefers-color-scheme as the default. Both are emitted in tokens.css.

## S3 Contrast — before and after

| Reference system, as found | | Corrected | |
|---|---|---|---|
| Secondary text<br>Specification dimensions, materials and tolerances | 3.81:1 | Secondary text<br>Specification dimensions, materials and tolerances | 7.17:1 |
| Tertiary / placeholder<br>Specification dimensions, materials and tolerances | 2.18:1 | Tertiary / placeholder<br>Specification dimensions, materials and tolerances | 4.65:1 |
| Placeholder on tint<br>Specification dimensions, materials and tolerances | 1.93:1 | Placeholder on tint<br>Specification dimensions, materials and tolerances | 4.11:1 |
| Destructive action<br>Specification dimensions, materials and tolerances | 3.27:1 | Destructive action<br>Specification dimensions, materials and tolerances | 6.54:1 |
| Sidebar section label<br>Specification dimensions, materials and tolerances | 3.67:1 | Sidebar section label<br>Specification dimensions, materials and tolerances | 7.98:1 |
| Control border<br>Specification dimensions, materials and tolerances | 1.58:1 | Control border<br>Specification dimensions, materials and tolerances | 3.05:1 |

**7 of 12 audited pairings in the reference system fail WCAG 2.1 AA.**

All 68 pairings in the new palette pass, in both light and dark mode. The full computed report ships as design-system/tokens/CONTRAST-REPORT.md and is regenerated from the token source, not maintained by hand.

1 Each row shows the same role at the reference system's value and at the corrected value, with the computed ratio.

2 The tonal step between primary, secondary and tertiary text is preserved — the ramp is darkened rather than flattened, so the hierarchy reads the same.

3 The destructive red moves from #FF4D4D to a darker tone. This is the most visible change in the palette, and it only affects text and icons; a solid red fill can stay bright.

4 The sidebar section label was the worst offender in the navigation at 3.67:1, because it sits at 8.5px as well as low contrast. It is corrected on both axes.

5 Ratios here are computed by the same function that generates the shipped report, so the document and the code cannot drift apart.

TYPE SCALE

# Specifications

| | | | | |
|---|---|---|---|---|
| | `display-2xl` | clamp(…) | 52–100px · 900 | Display |
| **Page title** | `display-xl` | 2.25rem | 36px · 800 | Display |
| **Specification title** | `display-lg` | 1.75rem | 28px · 800 | Display |
| **128** | `display-md` | 1.5rem | 24px · 900 | Display |
| **Section heading** | `display-sm` | 1.25rem | 20px · 700 | Display |
| Lead paragraph and specification description | `body-lg` | 1rem | 16px · 400 | Body |
| Base body copy, controls and navigation | `body-md` | 0.875rem | 14px · 400 | Body |
| Dense table rows, help text and metadata | `body-sm` | 0.8125rem | 13px · 400 | Body |
| **TECHNICAL SPECIFICATIONS** | `label-lg` | 0.8125rem | 13px · 700 | Display |
| **TABLE HEADER** | `label-md` | 0.75rem | 12px · 700 | Display |
| **SIDEBAR LABEL** | `label-sm` | 0.6875rem | 11px · 700 | Display |
| PI-000 · 150 × 400 mm | `mono-md` | 0.8125rem | 13px · 500 | Mono |

SECTION LABEL WITH FLEXING RULE · RETAINED FROM THE REFERENCE SYSTEM

**TECHNICAL SPECIFICATIONS**

**REVISION HISTORY**

SPECIFICATION ROW · THE MOST CONTENT-BEARING PATTERN

- Overall size: Width — × Height — [CODE]
- Portrait — printed both sides. Bleed: minimum —

1 Two families, as in the reference system: a display face for headings, labels, numerals and chips, and a body face for copy, controls and navigation. A mono face is added for reference codes and dimensions, which the reference system sets in the body face.

2 The scale bottoms out at 11px, not 8px. label-sm replaces six separate roles that previously sat between 8px and 10px.

3 Base body size rises from 13px to 14px. Every step is expressed in rem so the scale responds to user font-size preferences.

4 Tracking is retained: tight negative tracking on display sizes, wide positive tracking on uppercase labels. This is what gives the interface its character and it survives unchanged.

5 Specimens render in Poppins and Lato as stand-ins. The specified stack is Wix Madefor Display and Montserrat, with system fallbacks declared in tokens.css.

## S5 Spacing, radii and elevation

### SPACING SCALE · 4px BASE

| | | |
|---|---|---|
| space-px | 1px | hairline offsets |
| space-0.5 | 2px | |
| space-1 | 4px | icon gap |
| space-1.5 | 6px | |
| space-2 | 8px | control padding |
| space-3 | 12px | row padding |
| space-4 | 16px | block gap |
| space-5 | 20px | |
| space-6 | 24px | section gap |
| space-8 | 32px | panel padding |
| space-10 | 40px | |
| space-12 | 48px | major section |
| space-16 | 64px | |
| space-20 | 80px | |
| space-24 | 96px | hero rhythm |

### RADII

| xs | sm | md | lg | xl | full |
|---|---|---|---|---|---|
| 4px | 6px | 8px | 10px | 14px | 999px |

Radius scales with the element: controls take xs–sm, containers md–lg, sheets take xl, chips take full.

### ELEVATION

**shadow-0**
Flat — default for cards and rows

**shadow-1**
Top bar, sticky headers

**shadow-2**
Popover, dropdown, tooltip

**shadow-3**
Off-canvas drawer, command palette

**shadow-4**
Modal sheet

1 A formal 4px scale replaces the reference system's informal one. Seventeen values were in use there; all but five were already multiples of four, so imposing the scale causes almost no visual drift.

2 Half-steps at 2px and 6px are kept because the reference system's control padding depends on them.

3 Radii are unchanged from the reference system's observed values. The relationship — radius scales with element size — was already consistent and is now stated as a rule.

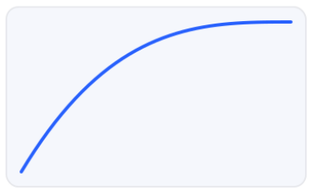
4 Elevation is five levels where the reference system had three ad-hoc shadows. Level 0 is the default: cards and rows are flat, separated by borders rather than shadows.

5 Shadows are tinted with the ink colour rather than pure black, so they stay tonally consistent with the palette.

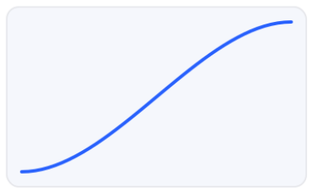
## S6 Motion, focus, breakpoints and layering

### MOTION

| | | |
|---|---|---|
| `--duration-instant` | `80ms` | Colour-only state change |
| `--duration-fast` | `140ms` | Hover, focus, button and nav states |
| `--duration-normal` | `220ms` | Popover, tooltip, badge change |
| `--duration-slow` | `300ms` | Route transition, accordion, sheet |
| `--duration-slower` | `420ms` | Drawer, media zoom |
| `--ease-expressive` | `cubic-bezier(.16, 1, .3, 1)` | Structural movement — panels, routes, accordions |
| `--ease-standard` | `cubic-bezier(.4, 0, .2, 1)` | State change — hover, focus, fills |
| `--ease-exit` | `cubic-bezier(.4, 0, 1, 1)` | Dismissal, collapse |



`ease-expressive`



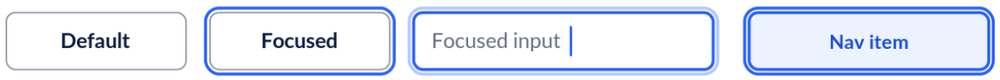
`ease-standard`

Two curves, as in the reference system.

Expressive for structural movement — routes, panels, accordions.
Standard for state change — hover, focus, fills.

**A reduced-motion query collapses every duration to 0.01ms.**
The reference system has no such query.

### FOCUS · A DESIGN-SYSTEM PRIMITIVE, NOT A PER-COMPONENT PATCH




2px ring in --color-border-focus, 2px offset, applied through :focus-visible so it never appears on pointer interaction.

### BREAKPOINTS

| | | |
|---|---|---|
| `xs` | `0` | 390px reference — single column |
| `sm` | `480px` | Compact mobile — icon-only chrome, overflow menu |
| `md` | `768px` | Tablet — sidebar goes off-canvas, split rotates to stack |
| `lg` | `1024px` | Small desktop — sidebar returns, split-pane restored |
| `xl` | `1440px` | Desktop — panel widens to 58% (variant A) |
| `2xl` | `1728px` | Wide — container max-width caps |

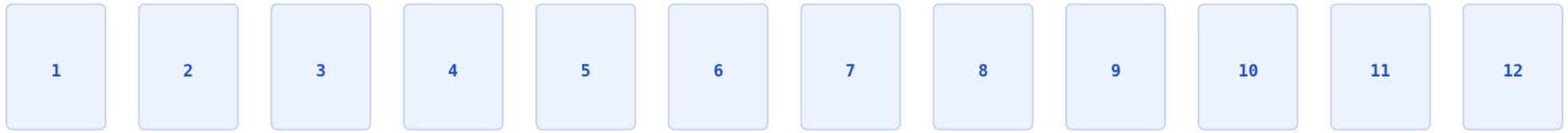
### Z-INDEX

| | |
|---|---|
| `base` | `0` |
| `raised` | `10` |
| `sticky` | `100` |
| `topbar` | `200` |
| `scrim` | `300` |
| `drawer` | `310` |
| `modal` | `400` |
| `popover` | `500` |
| `tooltip` | `550` |
| `toast` | `600` |

1 The two-curve vocabulary is retained exactly. It is the clearest piece of design thinking in the reference system.
2 Durations are formalised into five steps. The reference system clustered at 120–180ms and 280–350ms; those become fast and slow.
3 A prefers-reduced-motion query is added globally. This is a correction, not a refinement — the reference system plays every transition unconditionally.
4 Focus is a single primitive applied through :focus-visible. The reference system suppresses default outlines through border and background resets without replacing them, leaving keyboard focus invisible in parts of the interface.
5 Layering is formalised into ten named steps. The reference system used four ad-hoc values between 140 and 400.

## S7 Grid, containers and the shell

### 12-COLUMN FLUID GRID · 24px GUTTER

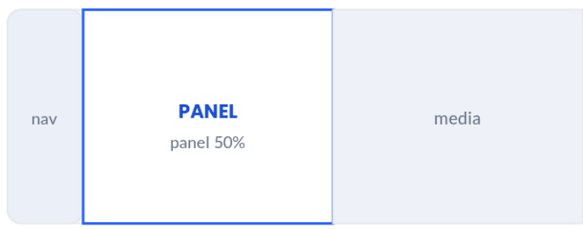



Twelve columns at lg and above · eight at md · four at sm and below. Gutter 24px, 20px, 16px respectively.

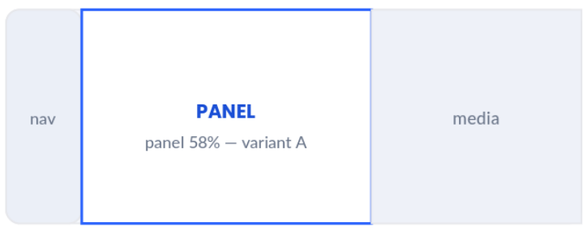
### CONTAINERS

| | |
|---|---|
| `prose` | 720px · Long-form text — notes, descriptions |
| `panel` | 640px · Specification data panel at its default 50% |
| `sheet` | 1020px · Editor sheet |
| `admin` | 960px · Management sheet |
| `page` | 1440px · Page content maximum |
| `full` | 1728 |

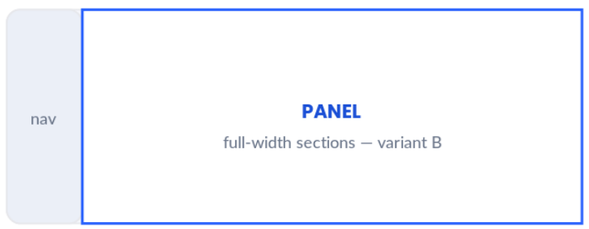
### SHELL · THE SPLIT AT ITS THREE STATES (VARIANT A + B, CONFIRMED)




`lg 1024–1439`




`xl 1440+`




`below the fold`

1 A twelve-column fluid grid governs the browse and landing surfaces. The specification detail view is not on the grid — it is a two-pane split, which is why the reference system needed no grid at all.

2 Container widths are named by purpose rather than size, so a component asks for the prose container rather than for 720px.

3 The prose container caps long-form notes at 720px, which is roughly 75 characters at body-md — the readable measure the reference system has no mechanism to enforce.

4 Variant A: the panel widens from 50% to 58% at 1440px, using space the media pane does not need.

5 Variant B: attachments, revision history and related specifications render full-width beneath the split, so the split governs only the identity-and-technical fold.

## 2.8 Token reference

### Colour — light

| Token | Value | Token | Value |
|---|---|---|---|
| --color-bg | #FFFFFF | --color-brand-hover | #16294C |
| --color-bg-subtle | #F5F7FC | --color-brand-subtle | #EBEFF7 |
| --color-bg-muted | #EEF1F8 | --color-brand-surface | #1B3769 |
| --color-bg-strong | #E3E8F4 | --color-brand-surface-hover | #22437C |
| --color-bg-inverse | #16294C | --color-brand-on-surface | #FFFFFF |
| --color-fg | #0D1E3F | --color-brand-on-surface-muted | #C3D0E8 |
| --color-fg-muted | #4B5870 | --color-info-fg | #2861FD |
| --color-fg-subtle | #6A7589 | --color-info-bg | #EDF2FF |
| --color-fg-on-accent | #FFFFFF | --color-info-border | #B9CCFF |
| --color-fg-on-inverse | #FFFFFF | --color-success-fg | #0F7A4D |
| --color-border | #E7E8EC | --color-success-bg | #E8F6EF |
| --color-border-strong | #CFD2D9 | --color-success-border | #A8DCC4 |
| --color-border-control | #8C94A4 | --color-warning-fg | #8A5A00 |
| --color-border-focus | #2861FD | --color-warning-bg | #FCF2DF |
| --color-accent | #2861FD | --color-warning-border | #EBD199 |
| --color-accent-hover | #1A4FD6 | --color-danger-fg | #B3261E |
| --color-accent-active | #1540B0 | --color-danger-bg | #FDECEA |
| --color-accent-subtle | #EDF2FF | --color-danger-border | #F2B8B4 |
| --color-accent-border | #B9CCFF | --color-neutral-fg | #42506B |
| --color-accent-fg | #1A4FD6 | --color-neutral-bg | #F0F2F7 |
| --color-brand | #1B3769 | --color-neutral-border | #CDD4E2 |

### Colour — dark

| Token | Value | Token | Value |
|---|---|---|---|
| --color-bg | #0C1424 | --color-brand-hover | #E2EAF8 |
| --color-bg-subtle | #121B2E | --color-brand-subtle | #16243F |
| --color-bg-muted | #18243C | --color-brand-surface | #111B2E |
| --color-bg-strong | #1F2D4A | --color-brand-surface-hover | #1A2842 |
| --color-bg-inverse | #E9EEF9 | --color-brand-on-surface | #E9EEF9 |
| --color-fg | #E9EEF9 | --color-brand-on-surface-muted | #A3B0C9 |
| --color-fg-muted | #9CA2AF | --color-info-fg | #8FB2FF |
| --color-fg-subtle | #7A808E | --color-info-bg | #14213A |
| --color-fg-on-accent | #08111F | --color-info-border | #2E4A80 |
| --color-fg-on-inverse | #0C1424 | --color-success-fg | #63C79B |
| --color-border | #2B3342 | --color-success-bg | #0E2620 |
| --color-border-strong | #454D5B | --color-success-border | #1F5B44 |
| --color-border-control | #5C6371 | --color-warning-fg | #E0B25C |
| --color-border-focus | #7FA6FF | --color-warning-bg | #251E0E |
| --color-accent | #7FA6FF | --color-warning-border | #5E4718 |
| --color-accent-hover | #9CBAFF | --color-danger-fg | #F08279 |

| Token | Value | Token | Value |
|---|---|---|---|
| --color-accent-active | #B7CDFF | --color-danger-bg | #2A1414 |
| --color-accent-subtle | #16243F | --color-danger-border | #6B2B26 |
| --color-accent-border | #2E4A80 | --color-neutral-fg | #A8B4CC |
| --color-accent-fg | #9CBAFF | --color-neutral-bg | #18202F |
| --color-brand | #C9D6EE | --color-neutral-border | #39445C |

## 2.9 Scale reference

### Type

| Token | Size | px | Weight | Tracking | Leading | Family | Role |
|---|---|---|---|---|---|---|---|
| display-2xl | clamp(3.25rem, 8vw, 6.25rem) | 52–100 | 900 | -0.04em | 0.92 | Display | Landing headline |
| display-xl | 2.25rem | 36 | 800 | -0.03em | 1.1 | Display | Page title |
| display-lg | 1.75rem | 28 | 800 | -0.02em | 1.15 | Display | Specification title |
| display-md | 1.5rem | 24 | 900 | -0.03em | 1.0 | Display | Statistic numeral |
| display-sm | 1.25rem | 20 | 700 | -0.01em | 1.25 | Display | Section heading, sheet title |
| body-lg | 1rem | 16 | 400 | 0 | 1.6 | Body | Lead paragraph, description |
| body-md | 0.875rem | 14 | 400 | 0 | 1.6 | Body | Base — body copy, controls, nav |
| body-sm | 0.8125rem | 13 | 400 | 0 | 1.55 | Body | Dense rows, table cells, help text |
| label-lg | 0.8125rem | 13 | 700 | 0.08em | 1.2 | Display | Section label with rule |
| label-md | 0.75rem | 12 | 700 | 0.1em | 1.2 | Display | Table header, group label, chip |
| label-sm | 0.6875rem | 11 | 700 | 0.12em | 1.2 | Display | Smallest permitted — sidebar label |
| mono-md | 0.8125rem | 13 | 500 | 0 | 1.5 | Mono | Reference codes, dimensions, tokens |

### Spacing · radii · elevation · motion

| Spacing | px | Radius | px | Motion token | Value |
|---|---|---|---|---|---|
| space-0 | 0px | radius-xs | 4px | --duration-instant | 80ms |
| space-px | 1px | radius-sm | 6px | --duration-fast | 140ms |
| space-0.5 | 2px | radius-md | 8px | --duration-normal | 220ms |
| space-1 | 4px | radius-lg | 10px | --duration-slow | 300ms |
| space-1.5 | 6px | radius-xl | 14px | --duration-slower | 420ms |
| space-2 | 8px | radius-full | 999px | --ease-expressive | cubic-bezier(.16, 1, .3, 1) |
| space-3 | 12px | | | --ease-standard | cubic-bezier(.4, 0, .2, 1) |
| space-4 | 16px | | | --ease-exit | cubic-bezier(.4, 0, 1, 1) |
| space-5 | 20px | | | | |
| space-6 | 24px | | | | |
| space-8 | 32px | | | | |
| space-10 | 40px | | | | |
| space-12 | 48px | | | | |
| space-16 | 64px | | | | |
| space-20 | 80px | | | | |
| space-24 | 96px | | | | |

### Breakpoints · layering · containers

| Breakpoint | px | Behaviour | z-index | Value | Container | px |
|---|---|---|---|---|---|---|
| xs | 0 | 390px reference — single column | z-base | 0 | prose | 720px |
| sm | 480px | Compact mobile — icon-only chrome, overflow menu | z-raised | 10 | panel | 640px |
| md | 768px | Tablet — sidebar goes off-canvas, split rotates to stack | z-sticky | 100 | sheet | 1020px |

| Breakpoint | px | Behaviour | z-index | Value | Container | px |
|---|---|---|---|---|---|---|
| lg | 1024px | Small desktop — sidebar returns, split-pane restored | z-topbar | 200 | admin | 960px |
| xl | 1440px | Desktop — panel widens to 58% (variant A) | z-scrim | 300 | page | 1440px |
| 2xl | 1728px | Wide — container max-width caps | z-drawer | 310 | full | 1728px |
| | | | z-modal | 400 | | |
| | | | z-popover | 500 | | |
| | | | z-tooltip | 550 | | |
| | | | z-toast | 600 | | |

# 3. Component inventory

Deliverable 7. 59 components across seven groups. This table is the source of truth for the Phase 5 build — every component listed here becomes a directory under /components with its own types, stories and tests.

## Origin key

| Origin | Meaning | Count |
|---|---|---|
| Retained | Exists in the reference system and carries over essentially unchanged. | 6 |
| Extended | Exists in the reference system but gains variants, states or accessibility behaviour. | 8 |
| Consolidated | Several inconsistent treatments in the reference system merged into one component. | 1 |
| Rebuilt | Exists in the reference system but is re-implemented because its current form has a structural defect. | 8 |
| Derived | New, but built from an existing reference-system pattern so it looks native. | 1 |
| New | No precedent in the reference system. Required by the brief. | 35 |

## Primitive — 6 components

| Component | Variants | States | Key props | Accessibility obligation | Origin |
|---|---|---|---|---|---|
| **Button** | primary · brand · secondary · subtle · ghost · danger · danger-ghost | default · hover · active · focus · disabled · loading | variant, size, iconStart, iconEnd, loading, disabled, as | 44×44 touch target; aria-disabled not disabled when a reason exists | Consolidated |
| **IconButton** | same seven variants | default · hover · active · focus · disabled | icon, label (required), variant, size | Accessible name required by type; tooltip on hover and focus | New |
| **Link** | inline · standalone · quiet | default · hover · focus · visited | href, external, underline | External links announce the new context | New |
| **Icon** | 32 names · 16 / 20 / 24px | — | name, size, title | aria-hidden when decorative; title when meaningful | New |
| **Spinner** | sm · md · lg | — | size, delayMs (default 400) | role=status with a polite live region | New |
| **VisuallyHidden** | — | — | children | Carries skip links and screen-reader-only labels | New |

## Form — 11 components

| Component | Variants | States | Key props | Accessibility obligation | Origin |
|---|---|---|---|---|---|
| **TextField** | text · email · url · number · password | default · hover · focus · filled · error · disabled · readonly | label, placeholder, help, error, required, prefix, suffix | Label always associated; error via aria-describedby and aria-invalid | Extended |
| **Textarea** | fixed · auto-grow · line-per-item | same seven | rows, maxRows, hint | Line-per-item variant documents its ** sub-heading convention in help text | Extended |
| **Select** | native · searchable | same seven | options, groups, placeholder, onCreate | Native under the hood; the searchable variant is a listbox with aria-activedescendant | Extended |
| **Combobox** | single · multiple | same seven | options, onSearch, onCreate, loading | Full combobox pattern; results announce a count | New |
| **TagInput** | — | default · focus · error · disabled | value, onAdd, onRemove, suggestions, max | Each chip removable by keyboard; removal announced | New |
| **Checkbox** | single · group | unchecked · checked · indeterminate · focus · disabled | checked, indeterminate, label, description | Indeterminate drives the table select-all | New |
| **Radio** | group | unselected · selected · focus · disabled | options, name, orientation | Arrow-key roving within the group | New |
| **Switch** | — | off · on · focus · disabled | checked, label | role=switch with aria-checked; never the only signal | Extended |
| **DropZone** | media · attachment | idle · drag-over · uploading · error | accept, multiple, maxSize, onDrop | Keyboard-reachable file input beneath; progress announced politely | Extended |
| **Field** | — | — | label, help, error, required, children | The wrapper that guarantees label association and reserves the error row | New |
| **FormActions** | inline · pinned | — | primary, secondary, dirty | Pinned variant stays reachable on long forms; guards unsaved changes | New |

## Data display — 12 components

| Component | Variants | States | Key props | Accessibility obligation | Origin |
|---|---|---|---|---|---|
| **Badge** | soft · solid · 5 semantic tones | — | tone, dot, size | Dot or icon accompanies colour so tone is never colour-alone | Derived |
| **StatusBadge** | draft · review · published · deprecated · superseded | — | status, showLink | Superseded requires and announces its replacement link | New |

| Component | Variants | States | Key props | Accessibility obligation | Origin |
|---|---|---|---|---|---|
| **Tag** | static · removable | default · hover · focus | label, onRemove, href | Removable tags expose a labelled remove control | New |
| **CodeChip** | — | default · copyable | code, copyable | Copy action confirms through a toast | Retained |
| **Card** | specification · media · stat · boxed group | default · hover · selected · disabled | elevation, selected, as, href | One link target, one focus stop; no nested interactives | New |
| **SpecRow** | row · sub-heading · two-column | — | text, reference, level | The reference system's core content pattern, carried over unchanged | Retained |
| **DefinitionList** | single · two-column | — | items (dynamic key/value), emptyValue | Real dl/dt/dd so pairs are announced as pairs | New |
| **Table** | default · compact · stacked (below md) | loading · empty · error · populated | columns, rows, sort, selection, onSort, stickyHeader | thead, th scope, caption, aria-sort; stacked variant keeps semantics | Rebuilt |
| **RevisionTable** | — | empty · populated | revisions, onRestore | Empty state sits below the header, not in place of it | New |
| **AttachmentList** | grid · list | empty · populated · uploading | files, onRemove, kinds | File type and size announced with the name | New |
| **MediaFrame** | primary · thumbnail · rail | empty · loading · loaded · error | src, alt, ratio, zoomable | Aspect ratio reserved; alt required; lazy below the fold | Extended |
| **Stat** | — | — | value, label | Empty value renders an em dash, not zero | Retained |

## Navigation — 9 components

| Component | Variants | States | Key props | Accessibility obligation | Origin |
|---|---|---|---|---|---|
| **Sidebar** | docked · off-canvas | open · closed | sections, selectedId, onSelect, filterQuery | Closed off-canvas is unmounted and inert; focus trapped when open | Rebuilt |
| **NavTree** | category · sub-group · leaf | default · hover · selected · expanded · focus | nodes, expandedIds, selectedId | Roving tab index; aria-expanded; collapsed branches unmounted | Rebuilt |
| **TopBar** | desktop · compact | — | breadcrumbs, actions, onMenuToggle | Contains the skip-link target | Extended |
| **Breadcrumbs** | full · truncated · back-only | — | trail, maxItems | nav with an ordered list and aria-current on the last item | Rebuilt |
| **AnchorRail** | horizontal · scrollable | default · active | sections, activeId | Scroll-spy with aria-current; keyboard-reachable, never scroll-only | New |
| **Tabs** | underline · pill | default · active · focus · disabled | tabs, activeId, onChange | Full tabs pattern with arrow-key navigation | Extended |
| **Pagination** | numbered · load-more | — | page, pageCount, pageSize, onChange | Page changes announced; rows-per-page is a labelled control | New |
| **CommandPalette** | overlay · full-screen (below md) | closed · empty · searching · results · no-results | onSearch, recents, groups | Focus trapped, Escape restores, results announce a count | New |
| **SkipLink** | — | hidden · focused | targetId | First focusable element on every page | New |

## Overlay — 6 components

| Component | Variants | States | Key props | Accessibility obligation | Origin |
|---|---|---|---|---|---|
| **Modal** | dialog · confirm · destructive-confirm | closed · open | title, size, onClose, initialFocus | Focus trapped, scroll locked, Escape closes, focus restored | Rebuilt |
| **Sheet** | bottom (mobile) · side · full-bleed | closed · open · dragging | side, snapPoints | The mobile presentation of Modal — same semantics | New |
| **Popover** | menu · form · info | closed · open | trigger, placement, offset | Takes focus; Escape returns it to the trigger | New |
| **Tooltip** | — | hidden · visible | content, placement, delay | Opens on hover and focus; never interactive; never sole source | New |
| **Menu** | actions · overflow · select | closed · open · item-focused | items, onSelect | Full menu pattern with type-ahead and arrow keys | New |
| **Scrim** | — | hidden · visible | onDismiss, blur | aria-hidden; click and Escape both dismiss | Retained |

## Feedback — 6 components

| Component | Variants | States | Key props | Accessibility obligation | Origin |
|---|---|---|---|---|---|
| **Alert** | info · success · warning · danger | static · dismissible | tone, title, action, onDismiss | Icon plus leading bar so tone is never colour-alone | New |
| **Toast** | info · success · warning · danger | entering · visible · paused · leaving | tone, message, action, duration | Polite live region; pauses on hover and focus | New |
| **EmptyState** | no-data · filtered · no-results · no-access | — | title, description, action, icon | Always carries the action that resolves it | New |
| **ErrorState** | inline · page · boundary | — | title, message, reference, onRetry | Retry offered; reference surfaced for support | New |
| **Skeleton** | text · row · card · media | — | lines, width, ratio | Reserves resolved dimensions; shimmer respects reduced-motion | New |
| **ProgressBar** | determinate · indeterminate | — | value, label | role=progressbar with value text | New |

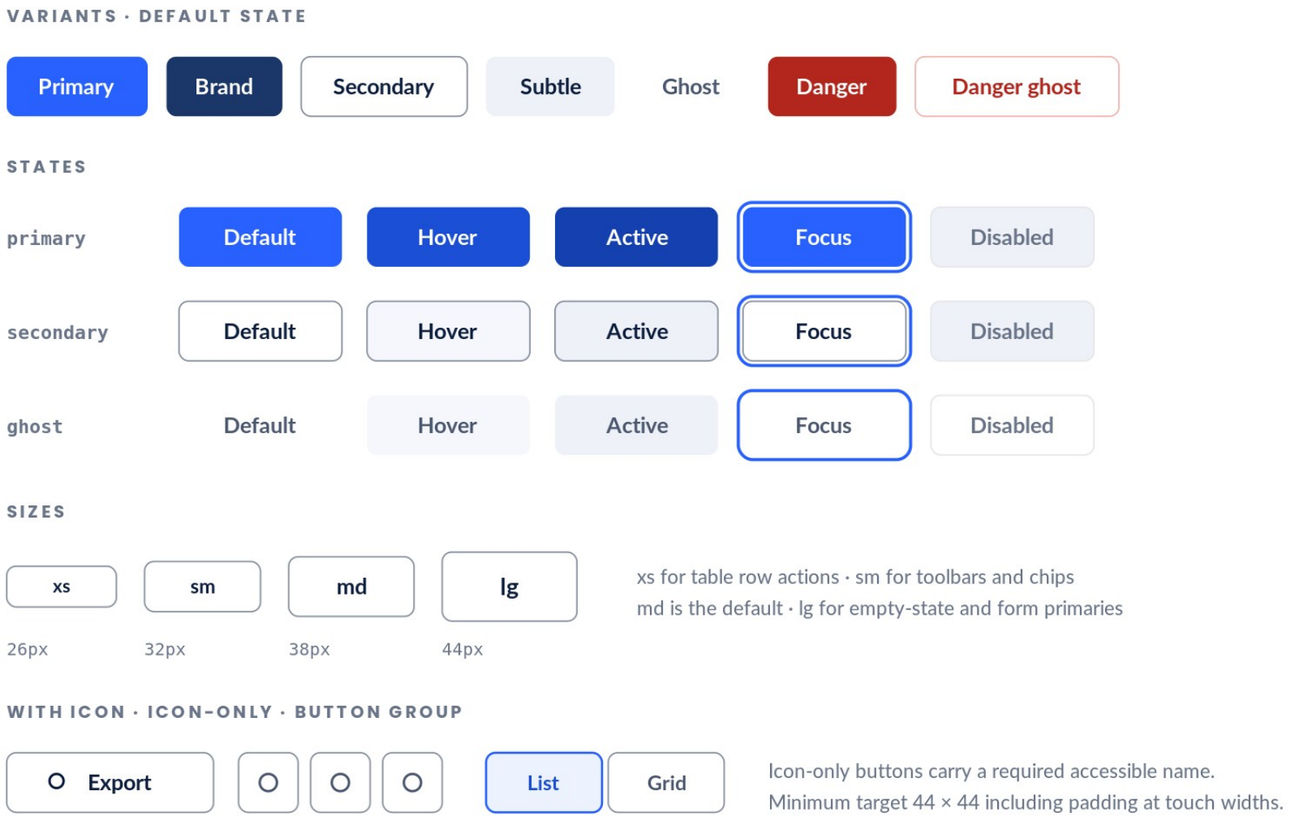
## Layout — 9 components

| Component | Variants | States | Key props | Accessibility obligation | Origin |
|---|---|---|---|---|---|
| **AppShell** | — | — | sidebar, topbar, children | Landmark regions: banner, navigation, main, contentinfo | Rebuilt |
| **SplitPane** | 50 / 50 · 58 / 42 (xl) · stacked (below lg) · rail-collapsed | — | panel, media, collapsed | Reading order is panel-first in the DOM at every width | Rebuilt |
| **Container** | prose · panel · sheet · admin · page · full | — | size, as | Caps measure at roughly 75 characters | New |
| **Grid** | 12 / 8 / 4 column | — | cols, gap, responsive | — | New |
| **Stack** | vertical · horizontal | — | gap, align, wrap | — | New |
| **SectionBlock** | — | — | label, children, anchorId | Renders the label-plus-flexing-rule header and registers with the anchor rail | Retained |
| **Accordion** | single · multiple | collapsed · expanded · focus | items, defaultExpanded | Measured height, no ceiling; collapsed content unmounted | Rebuilt |
| **DownloadBar** | — | empty · populated | files, actions | Pinned at every breakpoint; the one element that never moves | Retained |
| **FilterPanel** | rail · sheet (below md) | collapsed · expanded · active | groups, value, onChange, onClear | Active filter count announced; clear-all always reachable | New |

## 4. Components

Specimens for every component family, with the reasoning behind each decision. Variants, states and props are catalogued in Section 3.

## S8 Buttons




1 Seven variants. The reference system had four inconsistent treatments; these consolidate them and add the two danger forms it lacked.

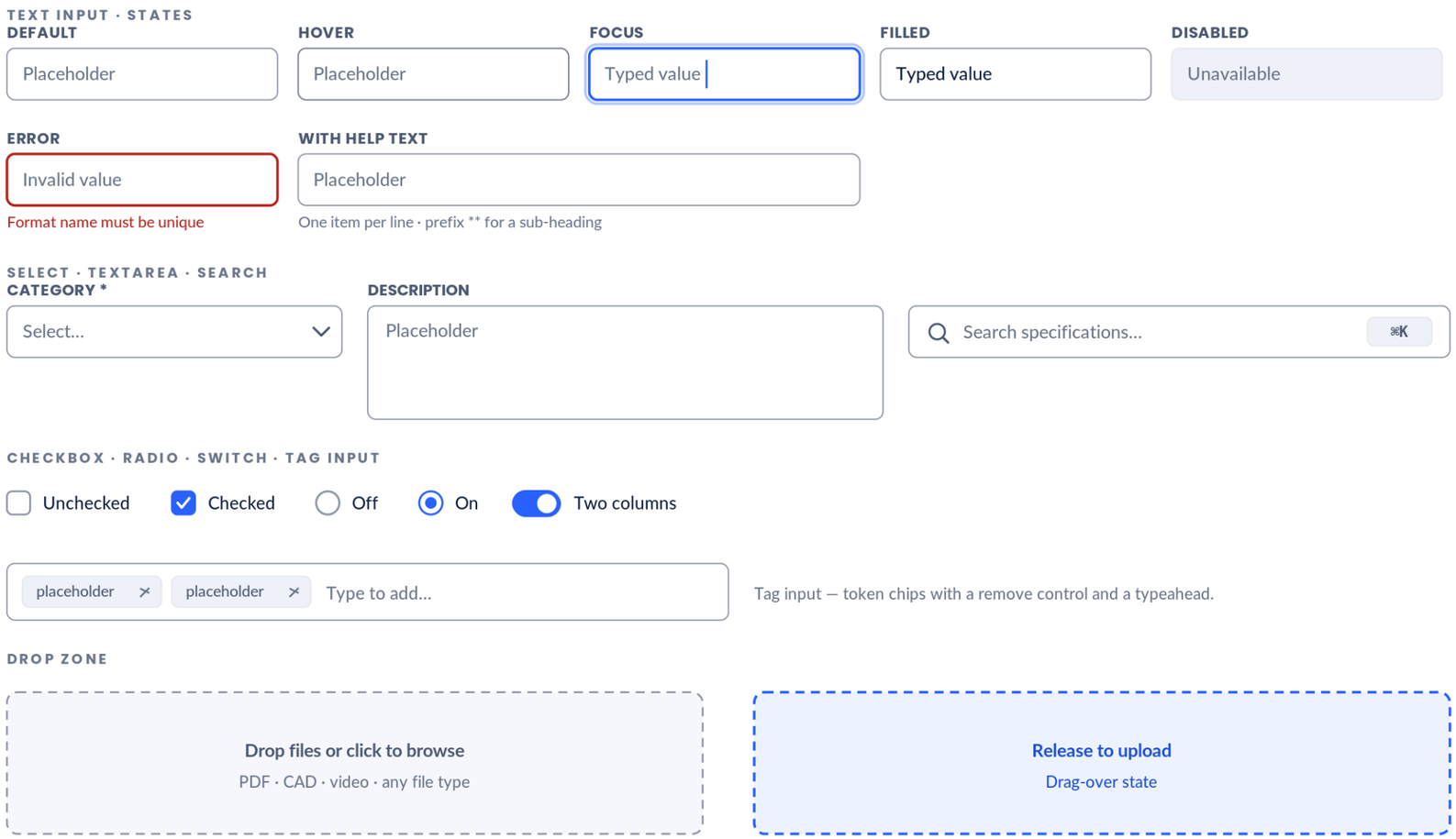
2 Five states on every variant, including the focus ring the reference system never defines.

3 Four sizes on a 26 / 32 / 38 / 44 ladder, derived from the control heights already in use — 26 for admin row actions, 32 for toolbars, 38 default, 44 for primary form actions.

4 Disabled uses a muted fill with subtle foreground and is never the only signal — disabled controls carry aria-disabled and a tooltip explaining why.

5 At touch widths every button reaches a 44 × 44 target through padding, without changing its visual height.

## S9 Inputs and form controls




1 Six states per field. The reference system defines default and focus only, with focus signalled by a border colour that does not reach 3:1.
2 border-control gives every field boundary a 3:1 minimum, satisfying WCAG 1.4.11 for non-text contrast.
3 Error state reserves its message row in the layout, so revealing an error never shifts the form.
4 Labels sit above inputs at every width. This keeps association unambiguous and touch targets full-width on mobile.
5 The tag input and drag-over state are net-new; the reference system has drop zones but no visible drag feedback.

## S10 Badges, status, tags and codes

### SPECIFICATION STATUS · SOFT (DEFAULT)

Draft · In review · Published · Deprecated · Superseded

### SOLID · FOR TINTED SURFACES

Draft · In review · Published · Deprecated · Superseded

### STATUS LIFECYCLE

Draft → In review → Published → Deprecated → Superseded

Deprecated and Superseded are terminal. Superseded carries a required link to the replacing specification; Deprecated does not. A specification cannot return to Draft once Published — it forks a new revision instead.

### REFERENCE CODE · TAG · COUNT · GENERIC BADGE

PI-000 · CODE-PLACEHOLDER · tag placeholder · removable × · 0 · INFO · SUCCESS · WARNING · DANGER · NEUTRAL

The reference code chip is the reference system's only badge. Every badge here is derived from it — same height, same radius, same display face at the same tracking — so the new elements read as native rather than imported.

### STATUS IN CONTEXT

**Specification title placeholder**

PI-000 · Published · —

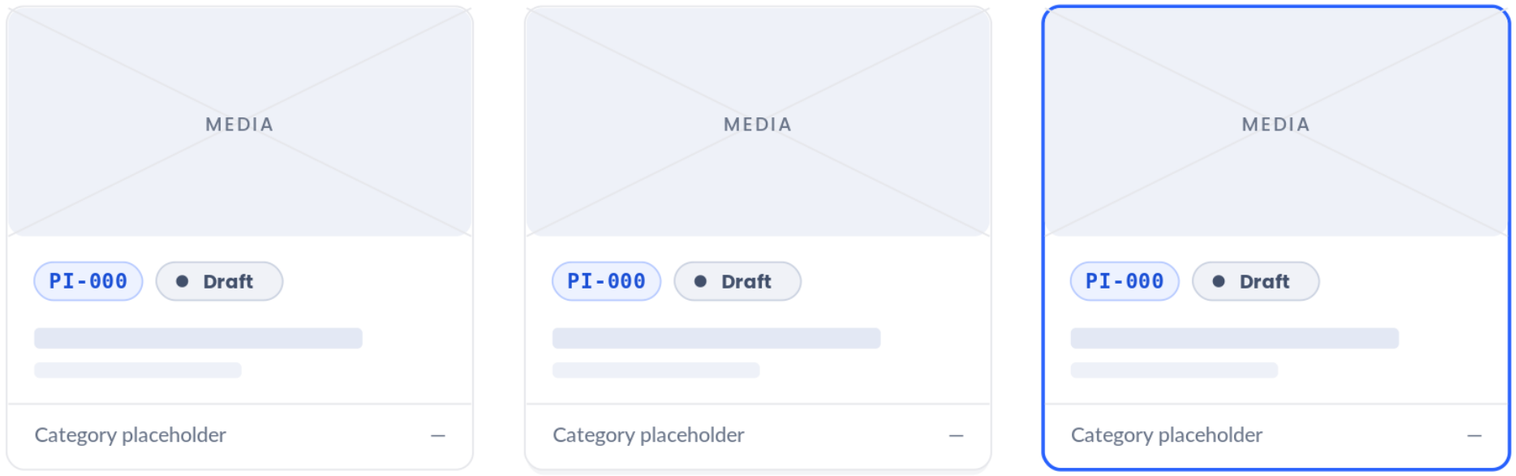
**Specification title placeholder**

PI-000 · Superseded · → replaced by —

1 The confirmed five-state vocabulary, rendered in both soft and solid forms.
2 A dot precedes each status label so state is not conveyed by colour alone — required by WCAG 1.4.1.
3 The lifecycle is enforced in the data layer, not just drawn. Terminal states cannot transition back to Draft.
4 Superseded requires a link to its replacement. This is the state that matters most for print specifications, where retired versions circulate physically.
5 Tags are visually distinct from status and from category — flat neutral chips, no dot, optionally removable. Three classification axes must never look alike.

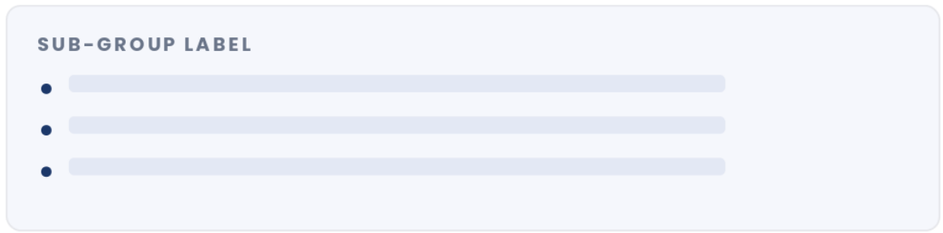
## S11 Cards

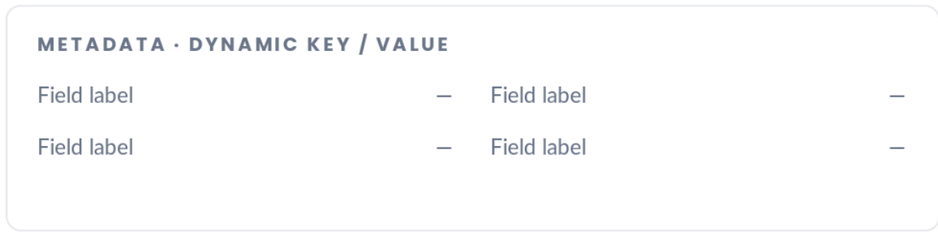
### SPECIFICATION CARD · GRID VIEW




**Default** **Hover** **Selected**

### BOXED SUB-GROUP · RETAINED FROM THE REFERENCE SYSTEM




1 The specification card has no precedent in the reference system, which has no grid, no summary card and no related-item card. It is built from the boxed sub-group's materials — same radius, same border, same tint.

2 Hover raises to shadow-1 and nothing else. No scale, no lift beyond 1px. Motion in this system is restrained by design.

3 Selected uses a 2px accent border rather than a fill, so the card's own content keeps its contrast.

4 The whole card is one link target and one focus stop. Nested interactive elements inside a card are prohibited.

5 Metadata renders as dynamic key/value rows — the confirmed structure. Labels are authored per specification or inherited from a category template, so a new field never needs a code change.

## S12 Tables

DATA TABLE · SEMANTIC, SORTABLE, SELECTABLE

| ☐ | NAME ^ | CODE | CATEGORY | STATUS | UPDATED ^ |
|---|---|---|---|---|---|
| ☐ | | PI-000 | | ● Superseded | — |

1 selected | Change status | Add tag | Archive

REVISION HISTORY · EMPTY, WITH THE COLUMN MODEL STILL LEGIBLE

| VERSION | DATE | AUTHOR | SUMMARY |
|---|---|---|---|

No revisions recorded

The empty state sits beneath the header, not in place of it, so the user can see what will appear here once the record is edited.

**Saving a specification writes a revision automatically.**

1 Real table semantics — thead, th with scope, and caption. The reference system draws its admin lists as CSS Grid rows, which screen readers cannot announce as rows and columns.

2 Sortable columns carry an indicator and aria-sort. Sorting does not exist in the reference system at all.

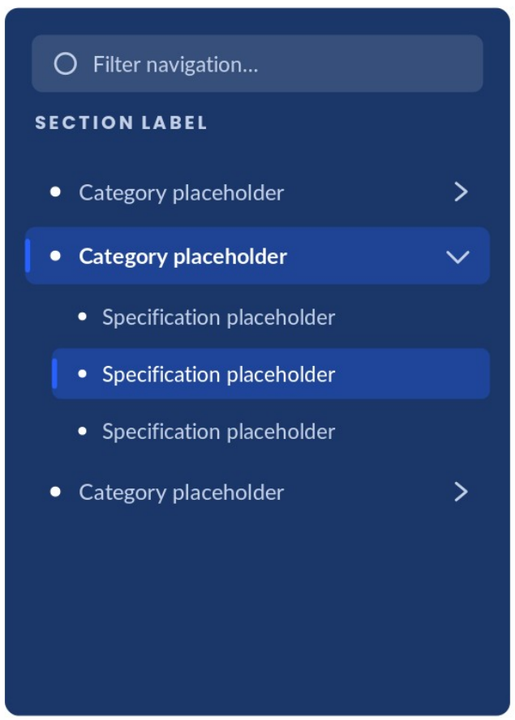
3 Selection column with an indeterminate header state, driving the bulk actions bar.

4 Row actions appear on hover and focus, and are always present in the DOM so keyboard users reach them.

5 Below 768px rows become stacked cards rather than scrolling horizontally — shown on Phase 2 plate R5.

## S13 Navigation, sidebar, breadcrumbs and tabs

SIDEBAR · TREE




BREADCRUMB

Home › Category › Sub-group › **Specification**

ANCHOR RAIL · SCROLL-SPY

Technical | Requirements | Metadata | Notes | Attachments

TABS

Specs | Categories | Users | Archive | Settings

PAGINATION

‹ 1 2 3 › Rows per page

TOP BAR

LOGO | Home › Category | Search specifications | Admin | Sign out

Selection = accent fill + 2px left edge bar. Retained exactly.

**Collapsed branches are unmounted, not hidden by overflow.**

1 The sidebar keeps the reference system's navy surface and its selection marker — accent fill plus a 2px left edge bar — because it is the clearest visual anchor in the product.

2 White-on-navy text is corrected: the section label moves from 45% to a solid tint that reaches AA, and rises from 8.5px to 11px.

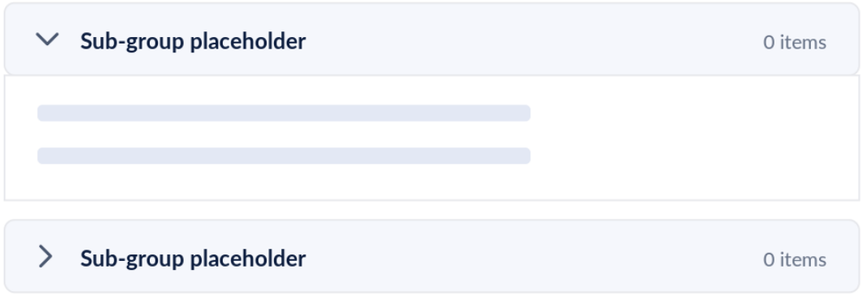
3 Collapsed branches are unmounted and the closed drawer is marked inert, so their controls leave the tab order. The reference system leaves them focusable while invisible.

4 Breadcrumbs become a real navigable trail with an ordered list and aria-current. The reference system's strip is decorative and drops its last segment on mobile.

5 The anchor rail is a scroll-spy segmented control — the mechanism that lets twelve sections live in a half-width panel.

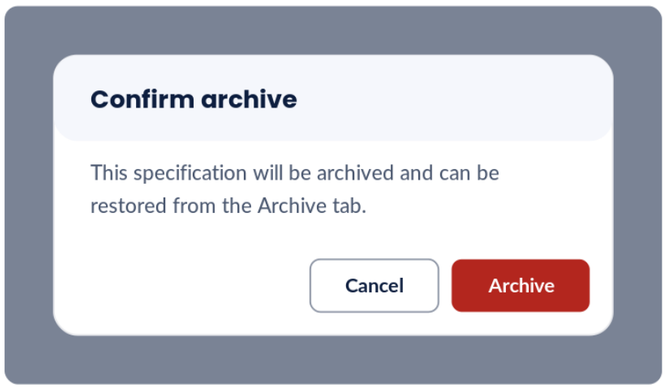
## S14 Accordions, modals, sheets, tooltips and popovers

ACCORDION

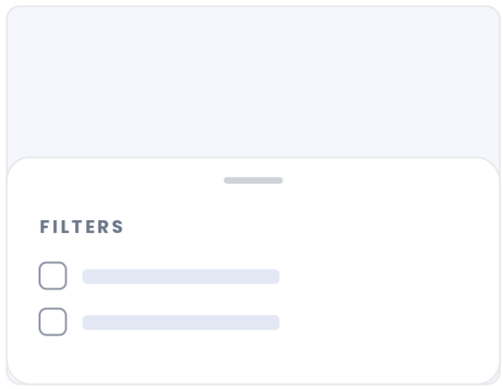



Height is measured, not capped. The reference system animates max-height to a fixed 600px ceiling, which **silently clips any branch taller than that.**

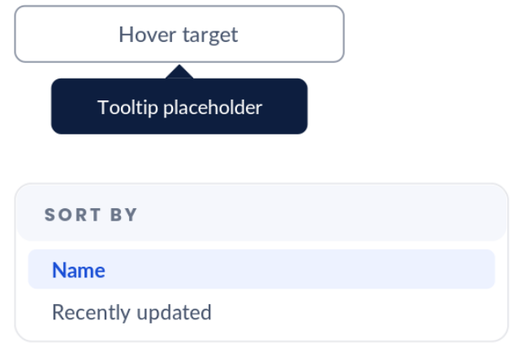
MODAL · SHEET · TOOLTIP · POPOVER




Modal — focus trapped, scroll locked, Escape closes, focus restored on close.




Bottom sheet — the mobile form of a modal.




Tooltip — describes, never the only source of information. Reachable on keyboard focus.

1 Accordion height is measured rather than capped, removing the reference system's 600px clipping ceiling.
2 Modals trap focus, lock background scroll, close on Escape and restore focus to the trigger. None of this is implemented in the reference system's overlays.
3 The bottom sheet is the mobile form of the modal — same component, different presentation, chosen by breakpoint rather than by a separate implementation.
4 Tooltips describe and never carry information available nowhere else. They open on hover and on keyboard focus, and never contain interactive content.
5 Popovers do contain interactive content and therefore take focus, unlike tooltips. Keeping the two distinct is what prevents keyboard traps.

## S15 Alerts and toasts

### ALERT · INLINE

**Information**
Placeholder message for a neutral notice.

**Saved**
Revision recorded automatically.

**Review required**
This specification has unpublished changes.

**Could not save**
Placeholder message with an error reference.

### TOAST

**Specification saved**
Placeholder detail
Undo

**Upload failed**
Placeholder detail
Retry

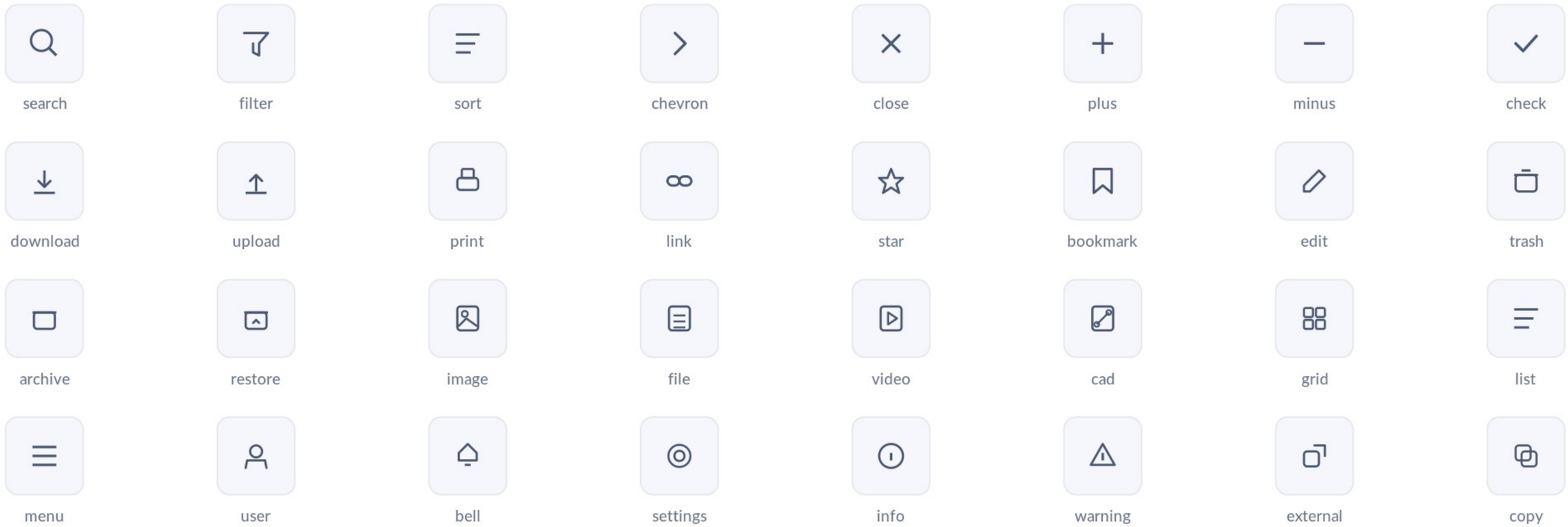
**Link copied**
Placeholder detail
Undo

Toasts announce through an aria-live polite region, auto-dismiss after 6 seconds, and pause on hover or focus.
A toast never carries the only route to an action — Undo is always also reachable from the Archive tab or the record itself.

1 Four semantic tones, each with an icon and a coloured leading bar so tone is never conveyed by colour alone.
2 Inline alerts sit in the layout and persist. Toasts float and expire. The two are never interchangeable.
3 Every toast announces politely to assistive technology and pauses its timer on hover or focus.
4 Destructive confirmations use a modal, never a toast — an action that cannot be undone must be confirmed before it happens, not offered an undo after.
5 The reference system has a notification panel but no inline alert and no toast; form errors surface only as browser validation.

## S16 Icons

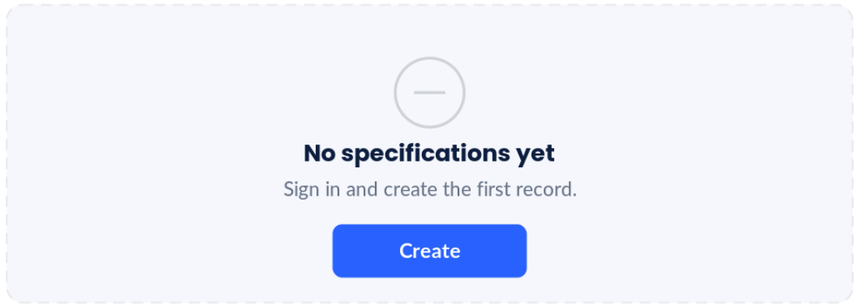
ICON SET · 20px GRID · 1.6px STROKE

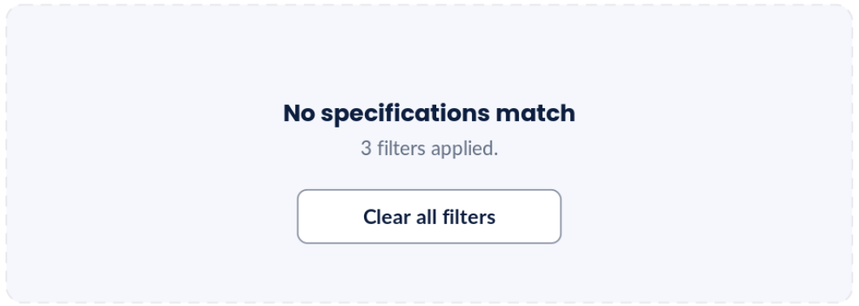



1 Thirty-two icons on a 20px grid with a 1.6px stroke, matching the reference system's stroke weight so they sit correctly beside its type.
2 A single typed Icon component with a required name prop. The reference system writes each icon out inline at its point of use, with no component and no sprite.
3 Decorative icons take aria-hidden. Icon-only buttons take an accessible name — the reference system's icon-only mobile controls have neither.
4 Icons never carry meaning alone. Every icon-only control has a tooltip, and every status icon is paired with a text label.
5 Sizes are 16, 20 and 24, tied to the type size they accompany rather than chosen per instance.

## S17 Empty, loading, error and skeleton states

### EMPTY · NO DATA AT ALL (THE SHIPPING STATE)




Filtered-empty is a distinct state — the resolution differs, so the message and the action differ.

### SKELETON · RESERVES FINAL DIMENSIONS



**Shown for any load expected to exceed 400ms.**

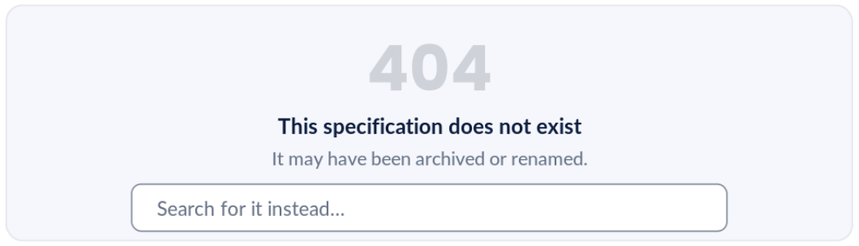
Dimensions match the resolved row exactly, so nothing shifts when data arrives. A shimmer animation is applied only when prefers-reduced-motion is not set.



Spinner — delayed 400ms so a fast response never flashes it.

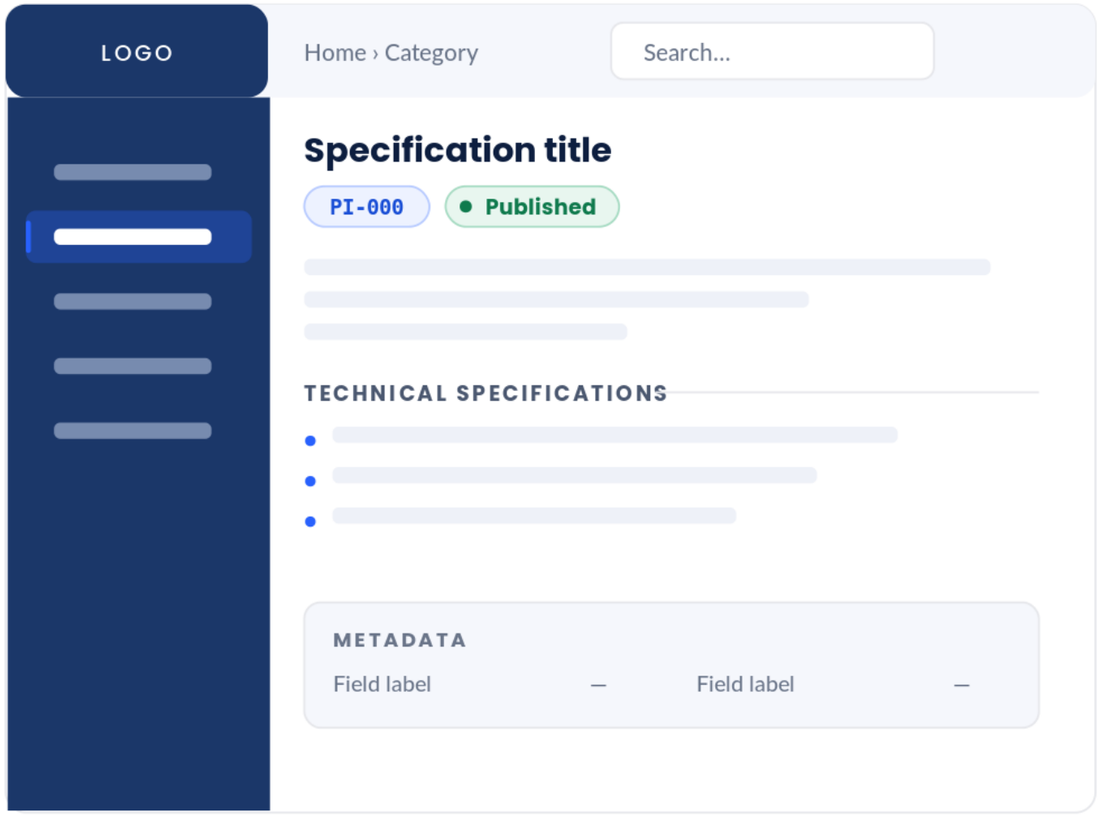
### ERROR · NOT FOUND



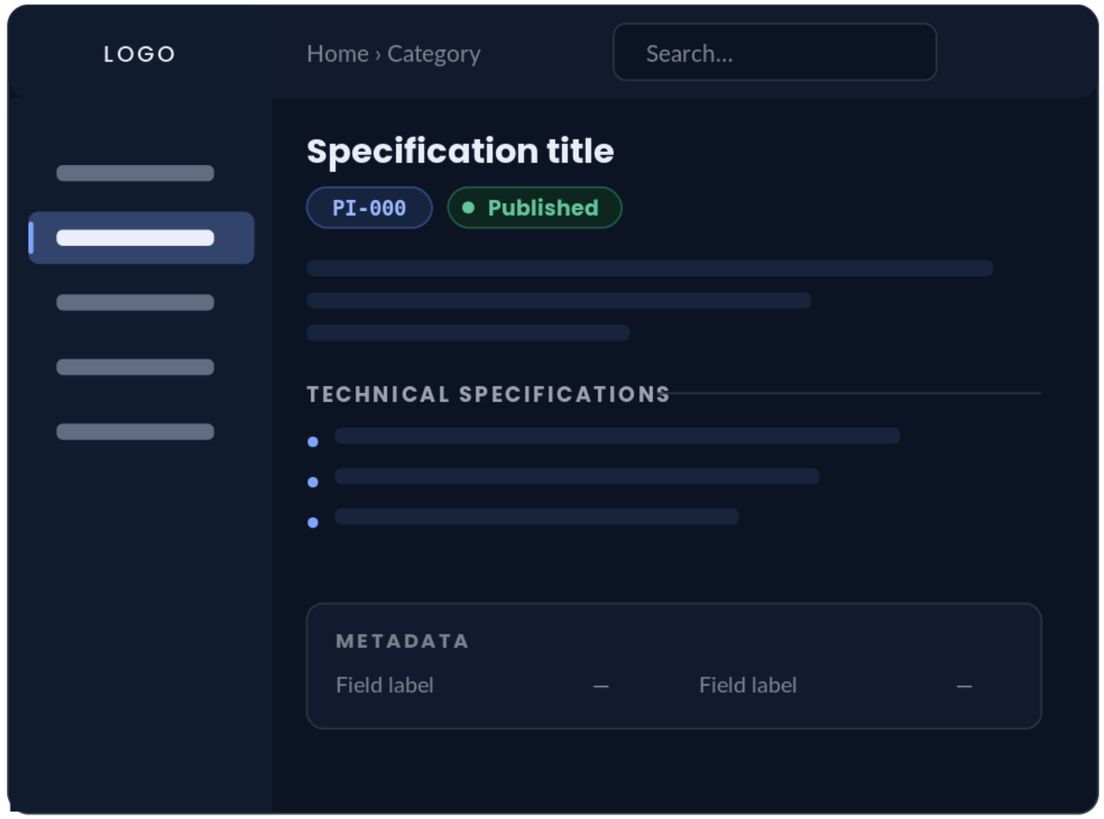




1 Because every specification ships empty, these are not edge cases — for a period they are the primary experience. The reference system defines none of them.
2 Empty-with-no-data and empty-after-filtering are separate components with different messages and different actions.
3 Skeletons reserve the resolved dimensions exactly, so data arrival causes no layout shift.
4 The spinner is delayed 400ms; the skeleton appears immediately. A fast response shows neither.
5 404 offers search rather than only a link home, because a dead specification link usually means the record was renamed rather than removed.

## S18 Theme applied — light and dark




**Light**




**Dark**

Identical markup. The only difference is the value bound to each token name — no dark variants, no duplicated components, no conditional classes. This is the payoff of naming tokens by role rather than by colour.

1 The same component tree rendered under both token sets. Nothing in the markup or the component code differs.

2 Dark mode is a Phase 9 requirement in your brief. Because it is derived from the token layer rather than retrofitted, it costs one stylesheet block.

3 The brand navy surface persists in dark mode as the sidebar, so the product's silhouette is recognisable in both themes.

4 Accent and semantic colours shift value but hold their hue, so a Published badge reads as the same thing in either theme.

5 Theme is chosen by a data-theme attribute with prefers-color-scheme as the fallback, and is exposed in the settings tab shown on Phase 2 plate D14.

## 5. Composition rules

Rules that govern how components combine. These are the ones that break systems when left unstated.

| Rule | Why |
|---|---|
| A card is one link target and one focus stop. Interactive elements are never nested inside it. | Nested interactives inside a link produce an invalid accessibility tree and make keyboard traversal unpredictable. |
| Only one primary button per view region. | Primary is a claim about the single most likely action. Two primaries is no primary. |
| Destructive actions are confirmed before they happen, never undone after. | An undo toast is not a substitute for confirmation on an irreversible action. Archive is soft, so it needs no modal; permanent delete is not, so it needs a typed confirmation. |
| Colour is never the only carrier of meaning. | Every status badge has a dot, every alert has an icon, every sort column has an arrow. WCAG 1.4.1. |
| Empty states carry the action that resolves them. | The catalogue ships empty. An empty state without a next step is a dead end at the exact moment the user needs direction. |
| Anything hidden is unmounted or inert, never merely invisible. | The reference system's collapsed branches and closed drawer stay in the tab order. This is the most common keyboard defect in the product. |
| Controls relocate on small screens; they are never removed. | The reference system deletes the image-download control below 360px. Function must not depend on viewport width. |
| Every field label sits above its input, at every breakpoint. | Keeps association unambiguous, keeps touch targets full-width, and removes an entire class of responsive layout bug. |
| Error message rows are reserved in the layout before an error exists. | Revealing an error must not shift the form under the user's cursor. |
| Skeletons match resolved dimensions exactly. | A skeleton that is the wrong size causes the layout shift it was introduced to prevent. |
| Tooltips describe; popovers contain. Tooltips never hold interactive content. | A tooltip with a button in it is unreachable by keyboard and is the classic source of focus traps. |
| Saving a specification always writes a revision. | History that depends on the author remembering to record it will not be maintained. |

## 6. Accessibility conformance

Target: WCAG 2.1 Level AA. Every Phase 1 finding is resolved at the system level rather than per component.

| Criterion | Obligation | How this system meets it |
|---|---|---|
| 1.4.1 Use of colour | Colour is not the sole means of conveying information | Status badges carry a dot, alerts carry an icon and a leading bar, sortable columns carry a direction arrow. |
| 1.4.3 Contrast (minimum) | Text reaches 4.5:1, large text 3:1 | Solved rather than chosen. All 74 pairings verified; report regenerated from source. |
| 1.4.4 Resize text | Text scales to 200% without loss of function | Every size is in rem against a 16px root. Type floor raised from 8px to 11px. |
| 1.4.10 Reflow | No horizontal scroll at 320px / 400% zoom | The shell scrolls internally without locking the document. The reference system sets overflow:hidden on html and body, which prevents reflow entirely. |
| 1.4.11 Non-text contrast | UI component boundaries reach 3:1 | border-control is a distinct token solved at 3.05:1, used on every input and secondary button. |
| 1.4.12 Text spacing | No loss of content when spacing is overridden | No fixed heights on text containers; the specification row and section label both flow. |
| 2.1.1 Keyboard | All functionality available from a keyboard | Command palette, roving tab index in the tree, full menu and tabs patterns, arrow-key groups. |
| 2.1.2 No keyboard trap | Focus can always leave a component | Tooltips hold no interactive content; modals trap deliberately and release on Escape. |
| 2.4.1 Bypass blocks | A mechanism to skip repeated content | SkipLink is the first focusable element on every page, targeting the main landmark. |
| 2.4.3 Focus order | Focus order preserves meaning | Collapsed accordions and the closed drawer are unmounted or inert, so they leave the tab order. |
| 2.4.7 Focus visible | Keyboard focus is visible | A single :focus-visible primitive — 2px ring, 2px offset. The reference system suppresses outlines without replacing them. |
| 2.5.5 Target size | Touch targets are at least 44×44 | Every control reaches 44×44 through padding at touch widths without changing its visual height. |
| 3.3.1 Error identification | Errors are identified in text | Field errors use aria-invalid and aria-describedby; the message row is reserved before it is needed. |
| 3.3.2 Labels or instructions | Inputs carry labels | The Field wrapper guarantees association. Placeholder is never the label. |
| 4.1.2 Name, role, value | Components expose correct semantics | Real tables, real dl for metadata, real dialog semantics, aria-current on breadcrumb and anchor rail. |
| 4.1.3 Status messages | Status is announced without focus change | Toasts and result counts announce through polite live regions. |

**REFERENCE SYSTEM AUDIT — THE MEASURED BASELINE**

Computed from the values as found in the published stylesheet. Every figure below is produced by the same function that generates the shipped contrast report.

| Pairing | Ratio | Required | Result |
|---|---|---|---|
| --txt on white (body) | 16.48:1 | 4.5:1 | PASS |
| --txt2 on white (secondary, 55%) | 3.8:1 | 4.5:1 | FAIL |
| --txt3 on white (tertiary, 35%) | 2.17:1 | 4.5:1 | FAIL |
| --acc on white (links, codes) | 4.98:1 | 4.5:1 | PASS |
| --navy on white (headings) | 11.69:1 | 4.5:1 | PASS |
| --txt2 on --bg1 (secondary on tint) | 3.52:1 | 4.5:1 | FAIL |
| --txt3 on --bg2 (placeholder on tint) | 1.92:1 | 4.5:1 | FAIL |
| nav section label, white 45% on navy | 3.67:1 | 4.5:1 | FAIL |
| nav sub-item, white 60% on navy | 5.28:1 | 4.5:1 | PASS |
| nav item, white 75% on navy | 7.31:1 | 4.5:1 | PASS |
| --red on white (destructive) | 3.27:1 | 4.5:1 | FAIL |
| --line3 border on white | 1.59:1 | 3:1 | FAIL |

# 7. Token delivery

The design system ships as files, not only as a document. These are in your project folder now, under design-system/tokens, and Phase 5 consumes them directly.

| File | Contains | Consumed by |
|---|---|---|
| tokens.css | Every token as a CSS custom property. Light set on :root, dark set on [data-theme="dark"] and under prefers-color-scheme. Includes the reduced-motion block and the :focus-visible primitive. | Imported once at the application root. |
| tailwind.tokens.js | Tailwind theme bridge. Colours, spacing, radii, shadows, type, screens, z-index and containers all resolve to the CSS variables above. | Spread into tailwind.config.js. Because values are variables, no dark: variants are needed anywhere. |
| tokens.ts | TypeScript literal unions for colour, type, spacing and radius tokens, plus the specification status list and breakpoint map. | Component prop types, so an invalid token is a compile error rather than a silent fallback. |
| CONTRAST-REPORT.md | The full computed verification — reference-system audit plus every pairing in both new palettes. | Regenerated from source. If a token changes and a pairing fails, the report says so. |

## 7.1 Why this matters for Phase 5

The reference system’s design decisions live inside an 80KB HTML file alongside its content, which is why they cannot be reused. Extracting them into typed, verified files is what turns this from a redesign of one site into the reusable platform your brief asks for. A future category, a future theme, or a future consumer of these tokens needs no code change.

## 8. Phase 4 scope & approval

On approval, Phase 4 produces the data model — deliverable 9. Reusable, fully typed, entirely empty.

### 8.1 Scope

- **Specification —** Identity, status, category and sub-group references, description, tags, technical specifications, requirements, dynamic metadata, notes, media, attachments, downloads, related specifications, revision history, timestamps and authorship.
- **Category —** Name, slug, description, order, media, optional sub-groups, and a metadata field template that new specifications inherit.
- **Sub-group —** Nullable middle tier. Name, slug, order, parent category.
- **Tag —** Name, slug, colour tone, usage count. Many-to-many, unordered, never navigable as a tree.
- **Status —** The five confirmed states with permitted transitions encoded, so the lifecycle is enforced rather than drawn.
- **Revision —** Version, timestamp, author, summary, and a diff of the changed fields.
- **Attachment / Download —** Filename, kind, size, MIME type, dimensions where applicable, storage key.
- **Metadata field —** Key, label, value, type, order. Dynamic per the confirmed decision.
- **User / Role —** The four roles named in the admin plates: Viewer, Editor, Approver, Administrator.

### 8.2 What Phase 4 will not do

It will not populate anything. Every model ships with empty values — empty strings, empty arrays, empty objects — exactly as your brief specifies. The seed data will be a single empty catalogue with no categories and no specifications.

### 8.3 Decisions needed

- Reference code format. The plates use PI-000 as a placeholder. Is the code authored, or generated from a category prefix and a sequence?
- Whether a specification can belong to more than one category. The current model says no — category is the exclusive primary axis and tags carry everything else. Confirm.
- Whether revision history records field-level diffs or only a summary. Field-level is more useful and more work.

### 8.4 Approval

If any foundation in Section 2 is wrong, it is far cheaper to change here than after fifty-nine components are built on top of it. The corrected palette in particular changes the smallest type sizes and the lightest text colours from the reference system — plate S3 shows exactly what that looks like side by side.

**End of Phase 3.** Deliverables 7–8 of 12 complete. Awaiting approval to proceed to Phase 4.